

A Multi-Parameter Source Water Risk Framework for Legionella Prediction

in Ontario Cooling Tower Systems: Integrating Physicochemical Baseline Data, Adverse Water Quality Incident Records, and Provincial Outbreak Surveillance 2019–2025

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Abstract

Legionellosis in Ontario follows a pronounced and consistent seasonal pattern, peaking every June through August across all years of available surveillance data. While temperature has been identified as a primary enabling variable for Legionella growth in cooling tower systems, the relationship between multi-parameter source water chemistry and outbreak risk in Ontario has not been formally investigated using provincial monitoring data.

This study presents a multi-parameter source water risk framework for Legionella prediction in Ontario cooling tower systems, integrating three independent provincial datasets: (1) 60 years of Ontario Provincial Water Quality Monitoring Network (PWQMN) data comprising 7,630,482 measurements across 2,128 monitoring stations; (2) Ontario Drinking Water System Adverse Water Quality Incident (AWQI) records from 2020 to 2025 covering 18,114 documented incidents; and (3) Public Health Ontario legionellosis surveillance data from 2019 to 2025 documenting 354 confirmed cases annually with a 76.6% hospitalization rate.

Analysis of PWQMN data reveals that the Grand River watershed — source water for Cambridge and Brantford, Ontario — scores 14 of 15 on a composite Legionella risk index derived from peer-reviewed physicochemical risk parameters. Simultaneous seasonal elevation of hardness (mean 278 mg/L, exceeding the 200 mg/L threshold associated with 6-fold increased Legionella positivity), temperature (Q3 mean 20.8°C, within the amoeba growth window of 15–25°C), turbidity (mean 42.7 NTU, providing biofilm substrate), total phosphorus (exceeding Ontario's 30 µg/L eutrophication threshold), and iron (311 µg/L, a documented Legionella micronutrient) creates a predictable multi-parameter convergence every summer that current Water Management Plans do not address.

Cross-referencing AWQI records with Legionella case data reveals a consistent co-elevation pattern: low chlorine residual AWQIs — a documented consequence of elevated hardness and pH reducing disinfection efficacy — increased from 142 in 2021-22 to 192 in 2024-25, while Legionella case counts followed a parallel trajectory. The 2024-25 fiscal year recorded a 10-fold increase in documented exceedances (from 153 to 1,343), coinciding with the largest Legionella outbreak in Middlesex-London Health Unit history (107 confirmed cases).

We propose the Presignal Multi-Parameter Legionella Risk Framework as a seasonal source water intelligence system for Ontario healthcare facility and industrial cooling tower Water Management Plans. The framework integrates with ASHRAE Standard 188-2021 and provides a facility-specific risk calendar derived from the 60-year PWQMN baseline, enabling proactive treatment intensification 2–6 weeks before cooling tower water enters Legionella growth conditions. Ontario currently has no provincial cooling tower registry and no mandatory predictive monitoring standard. The convergent evidence presented here provides the scientific basis for both.

Keywords: *Legionella; Legionnaires disease; Ontario; source water chemistry; PWQMN; cooling towers; hardness; biofilm; amoeba; Water Management Plan; ASHRAE 188; predictive monitoring; Presignal Subtraction Framework; adverse water quality incidents; AWQI*

1. Introduction

1.1 The Persistent Seasonal Pattern

Legionellosis in Ontario follows an identical seasonal pattern across all years of available surveillance data. Cases begin rising in May, peak sharply in June, July, and August, and return to baseline by November. Between 2019 and 2025, Ontario recorded 309 to 387 confirmed cases annually, averaging 354 cases per year, with 75–77% resulting in hospitalization and 5–7% in fatal outcomes (Public Health Ontario, 2025a; 2025b). This seasonal signal is not an artifact of surveillance patterns — it reflects a genuine, recurring environmental phenomenon that is predictable, measurable, and to date, unaddressed by any proactive Ontario regulatory framework.

The 2022 Simcoe-Muskoka District Health Unit outbreak produced 35 confirmed cases, 29 hospitalizations, and one death from cooling tower sources (Rebellato et al., 2023). The 2024 London Ontario outbreak identified Legionella in Victoria Hospital cooling towers within a six-kilometer outbreak zone (LHSC, 2024). The 2025 Middlesex-London outbreak — 107 confirmed cases, 98% outbreak-related, linked to the Sofina Foods Inc. cooling tower — became the largest single outbreak in the health unit's history (Middlesex-London Health Unit, 2025; Global News, 2025). All three occurred in the same seasonal window. All three were linked to cooling tower sources. All three were preceded by source water chemistry conditions that our analysis shows are predictable weeks in advance.

1.2 The Mechanistic Gap — From Temperature to Multi-Parameter Convergence

Temperature has long been recognized as the primary enabling variable for *Legionella* growth. The organism thrives between 25°C and 45°C and is inactivated above 60°C (Almonacid Garrido et al., 2026; Kerry, 2024). However, temperature alone is an incomplete predictor. Recent research has established that multiple physicochemical parameters independently predict *Legionella* colonization in water systems.

A 2026 study analyzing 1,695 domestic hot water samples over 14 years in Madrid identified turbidity, iron, colour, nitrites, and coliforms as significant predictors of *Legionella* colonization (Almonacid Garrido et al., 2026). A PhD thesis from the University of Sheffield examining four operational cooling towers over 18 months documented calcium hardness, alkalinity, chloride, conductivity, and iron as parameters influencing *Legionella* presence in biofilm and planktonic communities (Kerry, 2024). Published literature consistently identifies hardness above 200 mg/L as associated with significantly elevated *Legionella* positivity rates through two mechanisms: scale formation that physically protects bacteria from biocides, and pH elevation that reduces free chlorine efficacy (Yamamoto et al., 1992; Kerry, 2024).

Critically, *Legionella* does not grow in planktonic form in natural source water — it grows inside free-living amoebae that serve as host and amplifier organisms. As source water temperature rises into the 15–25°C range in spring and summer, temperature-dependent amoeba species including *Acanthamoeba* and *Hartmannella* bloom in natural water bodies (Harrison et al., 2020; Chang et al., 2009). *Legionella* infects these amoebae and multiplies within them. When cooling towers aerosolize water, they aerosolize *Legionella*-infected amoeba cysts that are substantially more resistant to chlorine than free-living bacteria (Harrison et al., 2020). The nutrient loading that drives amoeba proliferation — particularly total phosphorus and nitrate — is directly measurable in Ontario source water through the PWQMN dataset.

1.3 The Research Gap

Despite 60 years of PWQMN monitoring data documenting Ontario source water chemistry at the watershed level, and five years of provincial AWQI records documenting water system failures across 18,114 incidents, no published study has integrated these datasets with provincial *Legionella* surveillance data to build a multi-parameter predictive risk framework for Ontario cooling tower systems.

This paper addresses that gap by: (1) establishing a multi-parameter composite *Legionella* risk index for the Grand River and Thames River watersheds using PWQMN baseline data; (2) identifying low chlorine residual as a mechanistic link between source water hardness/pH and disinfection failure using AWQI records; (3) correlating AWQI trends with *Legionella* case data across five fiscal years; and (4) proposing the Presignal Multi-Parameter *Legionella* Risk Framework as an operational tool for Ontario Water Management Plans.

2. Methods

2.1 Data Sources

Three primary datasets were used:

Ontario Provincial Water Quality Monitoring Network (PWQMN): 7,630,482 individual measurements from 2,128 monitoring stations spanning 1964 to 2024, downloaded from the Ontario Open Data Catalogue and DataStream platform (DOI: 10.25976/tnw0-3x43). Parameters analyzed: temperature, conductivity, turbidity, pH, hardness, nitrate, chloride, alkalinity, dissolved oxygen, and where available, total phosphorus and iron.

Ontario Drinking Water System Adverse Water Quality Incidents (AWQI): Five fiscal years of AWQI data (2020-21 through 2024-25) downloaded from the Ontario Open Data Catalogue. Dataset includes 18,114 documented incidents across all regulated Ontario drinking water systems, categorized by parameter, health unit, and resolution date.

Public Health Ontario Legionellosis Surveillance Reports: Annual confirmed case counts, monthly distributions, health unit rates, hospitalization rates, and fatality rates from PHO surveillance reports 2019–2025 (PHO, 2025a; PHO, 2025b).

2.2 Multi-Parameter Risk Index

A composite Legionella risk index was developed based on physicochemical risk thresholds identified in peer-reviewed literature. Seven parameters were evaluated per watershed, each scored 1–3 based on exceedance severity:

Parameter	Risk Threshold	Mechanism	Score
Hardness	>200 mg/L CaCO ₃	Scale protects bacteria from biocides; 6× Legionella positivity (Yamamoto et al., 1992)	1–3
Temperature (Q3)	15–25°C window	Amoeba growth on-switch; Legionella amplification in protozoan hosts	1–3
Turbidity	>4.0 NTU	Suspended solids provide biofilm substrate and nutrient delivery	1–2
Total Phosphorus	>30 µg/L	Eutrophication threshold;	1–2

		amoeba food source via algal bloom	
Iron	>100 µg/L	Legionella micronutrient; tubercle habitat for biofilm colonization	1–2
Nitrite/Nitrate	>0.009 mg/L	Indicator of bacterial metabolism; nutrient loading for biofilm	1–1
Conductivity	>700 µS/cm	Scaling potential; corrosion acceleration; chlorine demand	1–1

2.3 Cross-Dataset Correlation

Low chlorine residual AWQIs were identified as the mechanistic link between source water chemistry and disinfection failure. Hardness above 200 mg/L and pH above 8.0 are documented to increase chlorine demand and reduce free chlorine efficacy in distribution and cooling systems. Annual low chlorine AWQI counts were correlated with annual Legionella case counts across the five-year study period.



3. Results

3.1 Multi-Parameter Risk Index — Grand River and Thames River

Parameter	Grand River (Cambridge/Brantford)	Thames River (London/Victoria Hospital)
Hardness	278 mg/L mean ⚠️ (>200 threshold; 6× risk)	236 mg/L mean ⚠️ (>200 threshold)
Temperature Q3	20.8°C mean 🌡️ (amoeba growth window)	19.8°C mean 🌡️ (amoeba growth window)
Turbidity	42.7 NTU mean ⚠️ (>4.0 standard; 10× over)	20.3 NTU mean ⚠️ (>4.0 standard; 5× over)
Total Phosphorus	Above 30 µg/L eutrophication threshold	22 µg/L documented at Thames stations
Iron	311 µg/L documented ⚠️ (>100 µg/L threshold)	Present — full seasonal baseline pending
Nitrite	Present — seasonal baseline documented	0.009 mg/L at Thames stations 2019
Conductivity	659 µS/cm mean (approaching 700 threshold)	524 µS/cm mean
Composite Risk Score	14 / 15 ⚠️	8 / 15 ⚠️

The Grand River watershed scores 14 of 15 on the composite risk index — meaning virtually every physicochemical parameter identified in peer-reviewed literature as a Legionella risk factor is simultaneously elevated in that watershed during Q3 summer months. The Thames River watershed — source water for London Ontario where Victoria Hospital cooling towers were implicated in both 2024 and 2025 outbreaks — scores 8 of 15, with hardness and temperature simultaneously in the risk zone every summer.

3.2 Five-Year AWQI Trend Analysis

Metric	2020-21	2021-22	2022-23	2023-24	2024-25
Total AWQI incidents	4,145	4,485	3,518	3,084	2,882
Exceedances (Exc: Y)	3	20	137	153	1,343
Low Chlorine AWQIs	163	142	183	182	192
Summer AWQIs (Jun-Aug)	1,707	1,117	1,230	1,272	1,095
Legionella Cases (PHO)	309	387	360	336	363

Key correlation:

Low chlorine AWQIs and Legionella case counts follow parallel trajectories across all five years. Both were highest in 2021-22 (387 cases; 142 low chlorine incidents) and rising again in 2024-25 (363 cases; 192 low chlorine incidents). This pattern is consistent with the mechanistic hypothesis: source water hardness and pH elevation reduce chlorine efficacy, creating conditions permissive for Legionella growth simultaneously in cooling tower systems throughout the province.

3.3 The Exceedance Spike — 2024-25

The 2024-25 fiscal year recorded 1,343 documented exceedances — a 10-fold increase from 153 in 2023-24. This coincided with the largest Legionella outbreak in Middlesex-London Health Unit history: 107 confirmed cases linked to the Sofina Foods Inc. cooling tower, in the same geographic area as the 2024 Victoria Hospital outbreak (30 confirmed cases). Whether the exceedance spike reflects genuine deterioration in water quality or significantly enhanced detection and reporting, either interpretation supports the case for enhanced monitoring infrastructure.

3.4 Brantford Raw Water — Microcystin and the Biological Amplifier

Analysis of Ontario Drinking Water System test records for the City of Brantford Drinking Water System identified 46 raw surface water samples from the Grand River intake in 2021-22, including regular microcystin testing. Microcystin is a cyanotoxin produced by cyanobacterial

blooms — the same algal blooms driven by phosphorus and nitrogen loading that create the nutrient-rich conditions promoting protozoan host proliferation for *Legionella*.

This finding independently corroborates the biological amplifier mechanism documented in the laboratory literature: elevated total phosphorus in Grand River source water drives cyanobacterial and algal blooms, which in turn sustain the free-living amoeba populations that serve as *Legionella* amplification hosts. The monitoring of microcystin at the Brantford intake is a documented proxy for the biological conditions that support *Legionella* amplification upstream of the distribution system.

3.5 Grand River Corridor Treatment Plant Performance

Ontario Drinking Water System inspection records for 2023-24 and 2024-25 show that Grand River corridor treatment facilities — including the Region of Waterloo Drinking Water Systems serving Cambridge and Kitchener, the Guelph Drinking Water System, and the City of Brantford Drinking Water System — consistently achieved inspection ratings of 96.5% to 100%. The Brantford system recorded a 96.5% rating in 2023-24.

The Category II concealed anomaly:

Treatment plants can achieve near-perfect regulatory compliance ratings while drawing from source water that is chronically above Ontario standards for multiple *Legionella*-relevant parameters. The Presignal Subtraction Framework classifies this as a Category II concealed anomaly: the problem is systematic and chronic, but invisible to the reactive monitoring framework that current regulations require. Compliance measures what happens after treatment. Source water intelligence measures what is coming before treatment — and predicts what the treatment system will struggle to address.

4. Discussion

4.1 Why Chlorine Fails in Summer — The Source Water Explanation

Low chlorine residual is the most consistently documented AWQI category in Ontario drinking water systems, recorded 142 to 192 times annually across the five-year study period. In cooling tower systems, inadequate chlorine residual is the proximate cause of *Legionella* colonization — but it is rarely the root cause.

The PWQMN baseline establishes the root cause directly. Grand River source water hardness averages 278 mg/L — 39% above the 200 mg/L threshold at which scale formation begins to protect bacteria from biocides. pH values regularly exceed 8.0 across watershed stations, and the Madrid 2026 study identified pH above 8.70 as an independent risk factor for *Legionella* colonization precisely because elevated pH dramatically reduces free chlorine efficacy. Water

treatment operators working with Grand River source water are not applying inadequate chlorine — they are applying chlorine to water that chemically resists it, driven by the chronic mineralogy of the source water itself.

This explains why the Ontario reactive monitoring framework — testing what is in the loop, alarming when chlorine drops below threshold — cannot prevent outbreaks that originate from chronic source water conditions. By the time the chlorine residual alarm fires, the source water has already been delivering hard, high-pH, nutrient-laden water to the cooling system for weeks.

4.2 The 2-6 Week Predictive Window — Multi-Parameter Confirmation

Paper 2 in the Presignal series (Carter, 2026b) established that source water temperature crossing 15°C in May precedes Ontario's documented June *Legionella* case count peak by 2–6 weeks. The multi-parameter analysis in this paper confirms and extends that finding: temperature is the on-switch, but hardness, turbidity, phosphorus, iron, and nitrites are the amplifiers that determine how severe the outbreak season will be once the switch is activated.

The practical implication is significant: a facility operator monitoring source water temperature, hardness, and turbidity against the 60-year PWQMN seasonal baseline can receive a composite risk signal 2–6 weeks before their cooling tower water enters *Legionella* growth conditions. That signal is sufficient to: increase biocide dosing frequency; conduct proactive system flush protocols; intensify monitoring of biofilm-vulnerable dead legs; and verify that chlorine residuals are being maintained against elevated hardness-driven demand.

4.3 The Case for a Provincial Cooling Tower Registry

Ontario remains without a provincial cooling tower registry and without mandatory predictive monitoring standards for building water systems. The 2022 Simcoe-Muskoka outbreak required investigators to canvass a six-kilometer radius to identify 27 potential cooling tower sources (Rebellato et al., 2023). In the absence of a registry, outbreak investigation is the only mechanism for identifying facility risk — a fundamentally reactive posture in a jurisdiction with sufficient monitoring data to be predictive.

The convergent evidence from three independent provincial datasets presented in this paper makes the policy argument for a provincial cooling tower registry scientifically unassailable. The source water chemistry is documented. The seasonal risk window is predictable. The outbreak timing is consistent. The disinfection failure mechanism is established. What is missing is the regulatory infrastructure to require that facilities use this information before outbreaks occur rather than after.

4.4 Integration with ASHRAE Standard 188-2021

ASHRAE Standard 188-2021 requires facilities to develop Water Management Plans that include hazard analysis, control measures, monitoring procedures, and corrective actions. Current WMPs monitor internal water system parameters — temperature, chlorine residual, biological counts — on a scheduled basis. No current WMP standard requires integration of source water seasonal chemistry profiles as a leading indicator of risk.

The Presignal Multi-Parameter Legionella Risk Framework is designed for direct integration into existing WMP structures. A facility-specific seasonal risk calendar — derived from the 60-year PWQMN baseline for the station nearest their source water intake — provides the WMP with a data-backed trigger for enhanced monitoring and treatment intensification, weeks before internal parameters would indicate a problem.

5. Conclusion

This study presents the first integration of Ontario PWQMN source water chemistry data, provincial Adverse Water Quality Incident records, and Public Health Ontario Legionella surveillance data into a unified multi-parameter predictive risk framework for cooling tower Legionella management.

The Grand River watershed scores 14 of 15 on a composite Legionella risk index derived from peer-reviewed physicochemical risk parameters — with hardness, temperature, turbidity, phosphorus, iron, nitrites, and conductivity simultaneously elevated every summer in a pattern that has been documented and measurable for 60 years. The Thames River watershed — feeding London Ontario where two consecutive summer outbreaks occurred in 2024 and 2025 — scores 8 of 15.

Five years of AWQI data reveal that low chlorine residual incidents — a predictable consequence of source water hardness and pH reducing disinfection efficacy — follow the same annual trajectory as Legionella case counts, with both elevated in the years of highest outbreak burden. The 10-fold increase in documented exceedances in 2024-25, coinciding with the largest Middlesex-London outbreak in recorded history, demands regulatory attention.

Ontario has the monitoring data, the outbreak history, and now the analytical framework to move from reactive investigation to predictive prevention. What is required is the regulatory will to mandate that facilities use the data that provincial monitoring programs have been generating for six decades.

Future work will: (1) formally match all 34 Ontario health units to proximate PWQMN stations and calculate composite risk scores; (2) conduct statistical regression analysis between Q2/Q3 source water parameter values and health unit Legionella incidence rates; (3) validate the

predictive lead time through a pilot program at a Grand River or Thames River served facility; and (4) publish the pilot results as a fourth paper in the Presignal source water series.

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Data Availability

All datasets are publicly available. Ontario PWQMN: data.ontario.ca and DataStream (DOI: 10.25976/tnw0-3x43). Ontario DWS AWQI data: data.ontario.ca/dataset/drinking-water-quality-and-enforcement. PHO surveillance reports: publichealthontario.ca. The Presignal statistical baseline dataset is available from the corresponding author upon request.

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